Hardware Architecture
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Sistema de Alerta de Proximidade com Controle de LED

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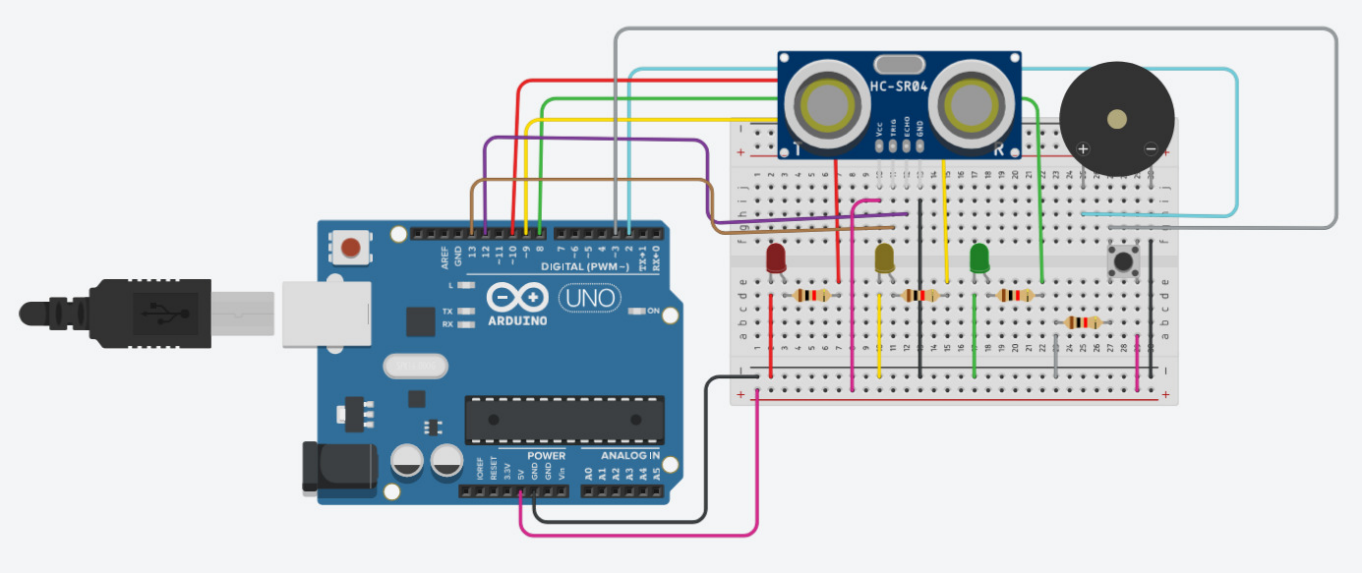
TinkerCad

Abaixo segue o projeto montado dentro da ferramenta TinkerCad.



Código

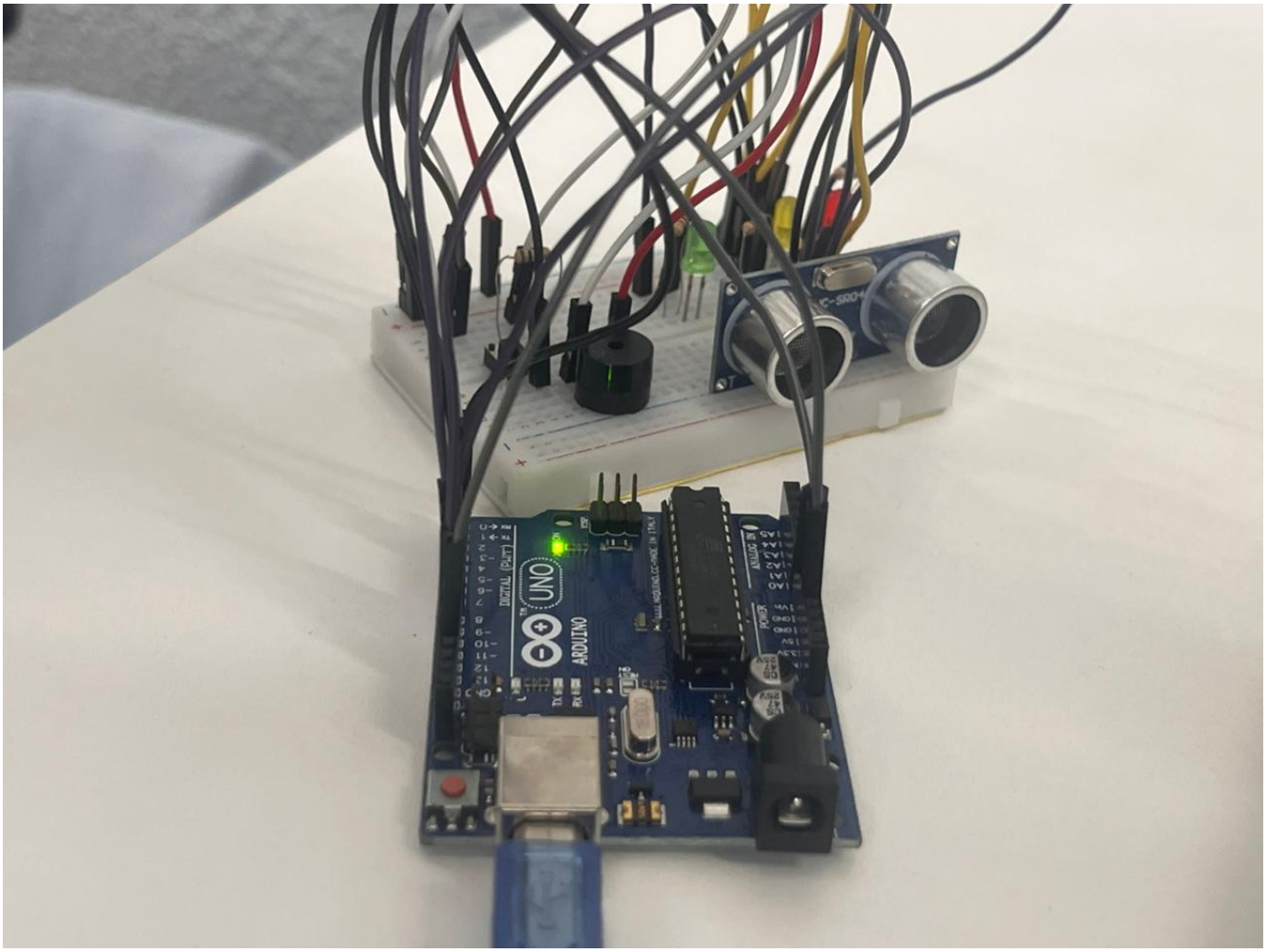
O circuito utilizou o seguinte código c++.

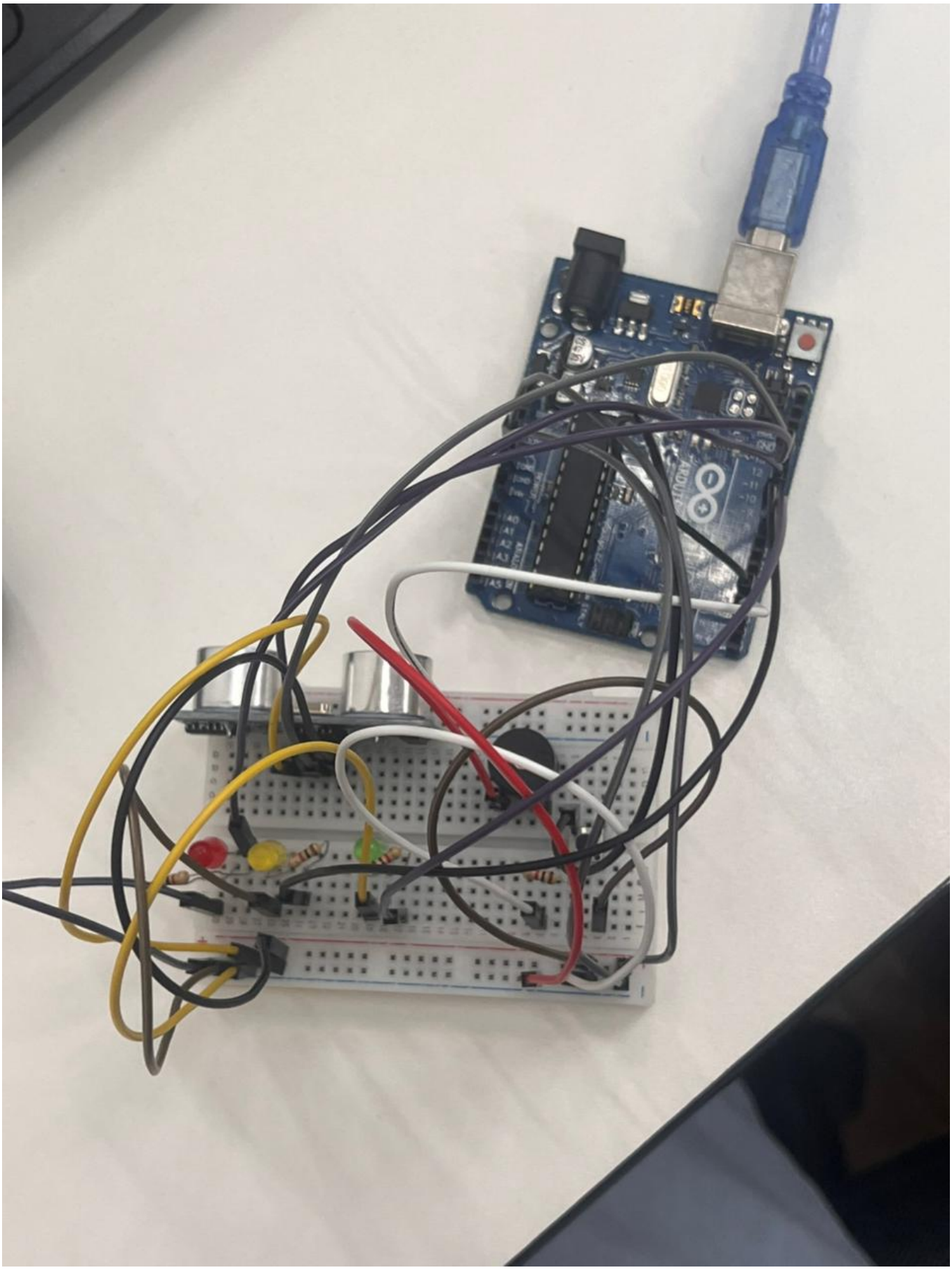
```
#define triggerPinUltra 13
#define echoPinUltra 12
#define redLed 10
#define greenLed 8
#define yellowLed 9
#define button 3
#define buzzer 2

bool systemActive = false;
bool lastButtonState = HIGH;

void setup() {
  delay(500);
  Serial.begin(9600);
  pinMode(redLed, OUTPUT);
  pinMode(greenLed, OUTPUT);
  pinMode(yellowLed, OUTPUT);

  pinMode(triggerPinUltra, OUTPUT);
  pinMode(echoPinUltra, INPUT);

  pinMode(buzzer, OUTPUT);

  pinMode(button, INPUT);
}

void loop() {
  bool buttonState = digitalRead(button);

  //verifica o ultimo estado do botao, para persistir
  if(buttonState == LOW && lastButtonState == HIGH){
    systemActive = !systemActive;
  }

  lastButtonState = buttonState;

  //se o botao esta LOW, fica em standby
  if(!systemActive){
    controlLeds(LOW, LOW, LOW);
    return;
  }

  //pega a distancia do ultrassonico
  long distance = getDistance(triggerPinUltra, echoPinUltra);

  lastButtonState = buttonState;

  //distancia media
```

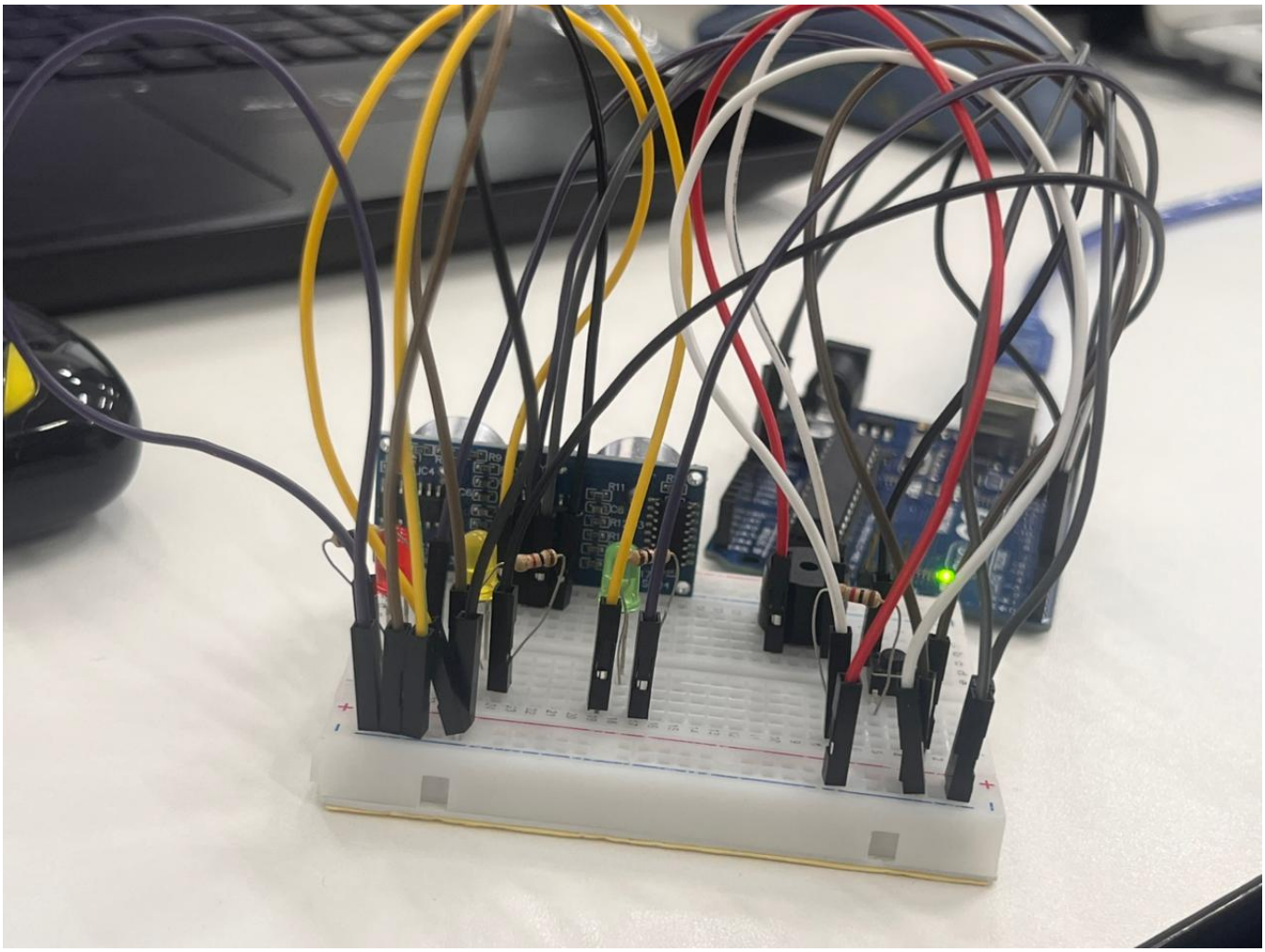
```
  if (distance <= 30 && distance >= 15) {
    analogWrite(buzzer, 0);
    controlLeds(LOW, HIGH, LOW);
    return;
  }

  //distancia perigo
  if (distance < 15) {
    controlLeds(HIGH, LOW, LOW);
    analogWrite(buzzer, 255);
    return;
  }

  //distancia ok
  analogWrite(buzzer, 0);
  controlLeds(LOW, LOW, HIGH);
}

void controlLeds(bool red, bool yellow, bool green){
  digitalWrite(redLed, red);
  digitalWrite(yellowLed, yellow);
  digitalWrite(greenLed, green);
}

long getDistance(int triggerPin, int echoPin) {
  long duration, distance;
  digitalWrite(triggerPin, LOW);
  delayMicroseconds(2);
  digitalWrite(triggerPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(triggerPin, LOW);
  duration = pulseIn(echoPin, HIGH);
  distance = (duration / 2) / 29.1;
  Serial.print("Distancia em cm: ");
  Serial.println(distance);
  delay(50);

  return distance;
}
```

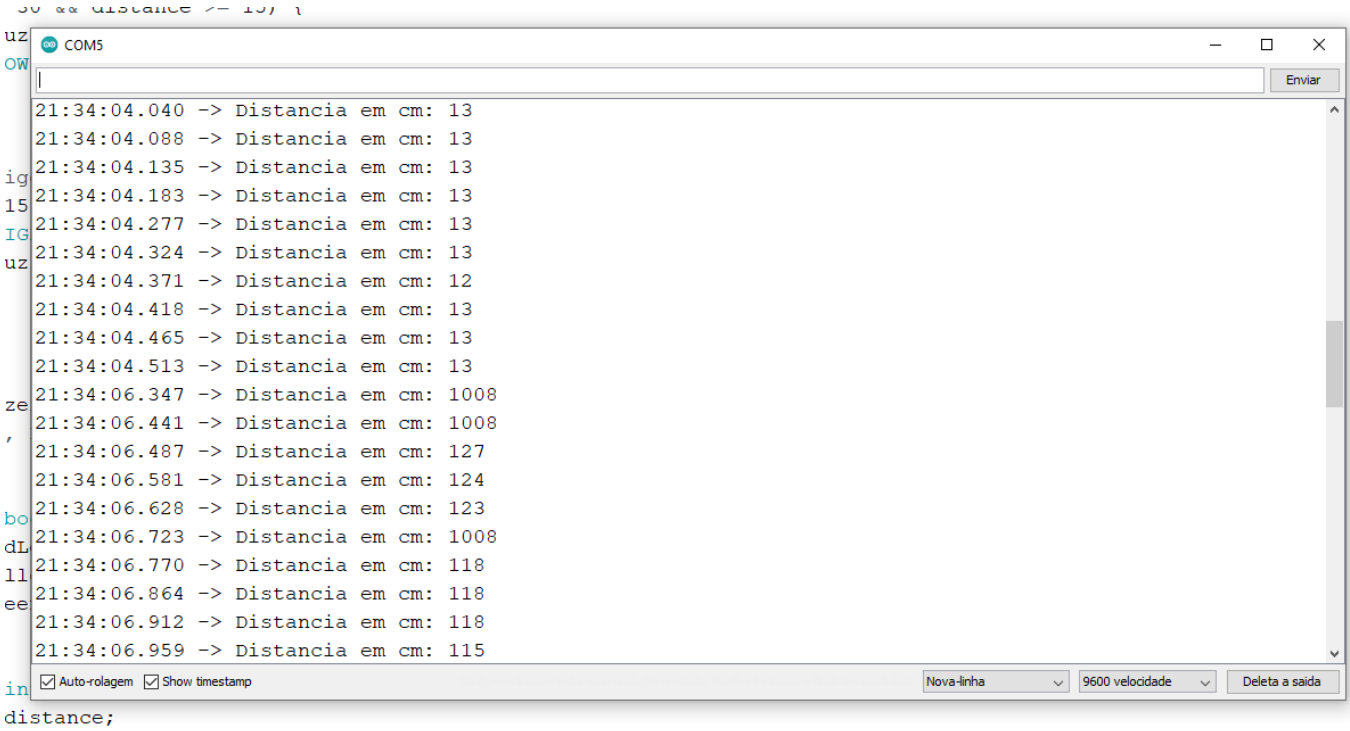
Projeto Pronto

Fotos do projeto montado e do serial marcando a distância pelo sensor ultrassônico.








COM5
Enviar
21:34:04.040 -> Distancia em cm: 13
21:34:04.088 -> Distancia em cm: 13
21:34:04.135 -> Distancia em cm: 13
21:34:04.183 -> Distancia em cm: 13
21:34:04.277 -> Distancia em cm: 13
21:34:04.324 -> Distancia em cm: 13
21:34:04.371 -> Distancia em cm: 12
21:34:04.418 -> Distancia em cm: 13
21:34:04.465 -> Distancia em cm: 13
21:34:04.513 -> Distancia em cm: 13
21:34:06.347 -> Distancia em cm: 1008
21:34:06.441 -> Distancia em cm: 1008
21:34:06.487 -> Distancia em cm: 127
21:34:06.581 -> Distancia em cm: 124
21:34:06.628 -> Distancia em cm: 123
21:34:06.723 -> Distancia em cm: 1008
21:34:06.770 -> Distancia em cm: 118
21:34:06.864 -> Distancia em cm: 118
21:34:06.912 -> Distancia em cm: 118
21:34:06.959 -> Distancia em cm: 115
Auto-rolagem
Show timestamp
Nova-linha
9600 velocidade
Deleta a saida